Quantitative Methods 1

Chapter 1: Organising and presenting data

Exercise worksheet

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Exercise 1 — A survey of retail outlets

A retail chain operates 180 shops. It randomly selects 60 shops from its complete list and records four characteristics for each: (i) the main channel through which sales are made (physical store, website or mobile application); (ii) the number of products returned over the same week; (iii) turnover during one working day (in euros); and (iv) service quality, rated low, medium or high. Each shop belongs to one main sales-channel category only.

1. Identify the population, sample and statistical unit.
2. Classify all four variables and choose a suitable graph for each.
3. For which variables do cumulative frequencies make sense?
4. Explain why random selection from the complete list is preferable to a voluntary-response survey. Give two potential biases associated with voluntary participation.

Exercise 2 — Product returns: discrete distribution

Weekly product returns were counted for the same 60 shops in Exercise 1:

Returns x_i	0	1	2	3	4	5
Shops n_i	5	11	19	14	8	3

1. Check the total number of shops.
2. Build the full table of counts, frequencies, ascending and descending cumulative counts and frequencies.
3. Draw a stick diagram.
4. Find the counts and percentages for at most two and at least three returns. Verify that these groups partition the sample.

Exercise 3 — Price survey: unequal class widths

A firm surveys 150 consumers about the maximum price each would be willing to pay for a portable solar charger. Prices are nonnegative and the recorded classes are left-closed and right-open: for example, EUR 36 belongs to $[36, 45[$, not $[30, 36[$.

Class (euros)	12–18	18–24	24–30	30–36	36–45	45–60	60–75
Count n_i	6	14	28	42	30	20	10

1. Check the total count; calculate each class width, relative frequency, percentage and count density $d_i = n_i/a_i$.
2. Construct a density-height histogram. Build the increasing and decreasing cumulative-count tables and plot both cumulative curves, indicating their vertices.
3. Find the shares whose declared maximum price is below EUR 36 and at least EUR 36. Infer the proportion able to buy at a price of EUR 36 if maximum willingness to pay is the only constraint.
4. Explain why raw counts cannot be used as histogram heights for unequal class widths. Check the areas of two bars.

General instructions: show formulas, intermediate calculations and units. Label axes and scales. Round percentages to two decimal places.